

ITEP 414 – System Administration and Maintenance

WEEK 3 PORTFOLIO PROJECT

Enterprise Server Deployment and Operating System Installation

Bachelor of Science in Information Technology (BSIT)

Week 3 • Individual Portfolio Project

Regie M. Estravela

1. Project Overview

This project documents the deployment of an Ubuntu Server virtual machine for a simulated startup enterprise environment. The server was installed, configured, secured for remote administration using SSH, verified through standard Linux commands, and prepared for professional documentation.

2. Learning Objectives

- Install Ubuntu Server in a virtual machine using recommended settings.
- Configure hostname, user account, storage, networking, and SSH.
- Verify server functionality using standard Linux commands.
- Explain the difference between BIOS and UEFI.
- Describe the Ubuntu boot process.
- Compare Windows Server, Ubuntu Server, and Rocky Linux.
- Create professional technical documentation for a server deployment.

3. Virtual Machine Specifications

Component	Configuration
VM Name	Ubuntu-Server-Week03
OS	Ubuntu Server 24.04.4 LTS
RAM	4 GB
CPU	2 virtual processors
Storage	40 GB VDI
Network	NAT / DHCP
Hostname	server01
User	regie (non-root administrative user)
SSH	OpenSSH Server

4. Ubuntu Server Installation Summary

Language and Keyboard

English was selected and the appropriate keyboard layout was configured.

Network

DHCP was accepted and the server received an IPv4 address.

Hostname

The hostname was configured as server01 as required by the module.

User Account

A non-root administrative user was created.

Storage

Guided – Use Entire Disk was selected for the 40 GB virtual disk. LVM encryption was not enabled.

SSH

Install OpenSSH Server was enabled for secure remote administration.

Additional Packages

No additional snaps were selected.

Installation Completion

The installation finished successfully and the Ubuntu installation ISO was removed before reboot.

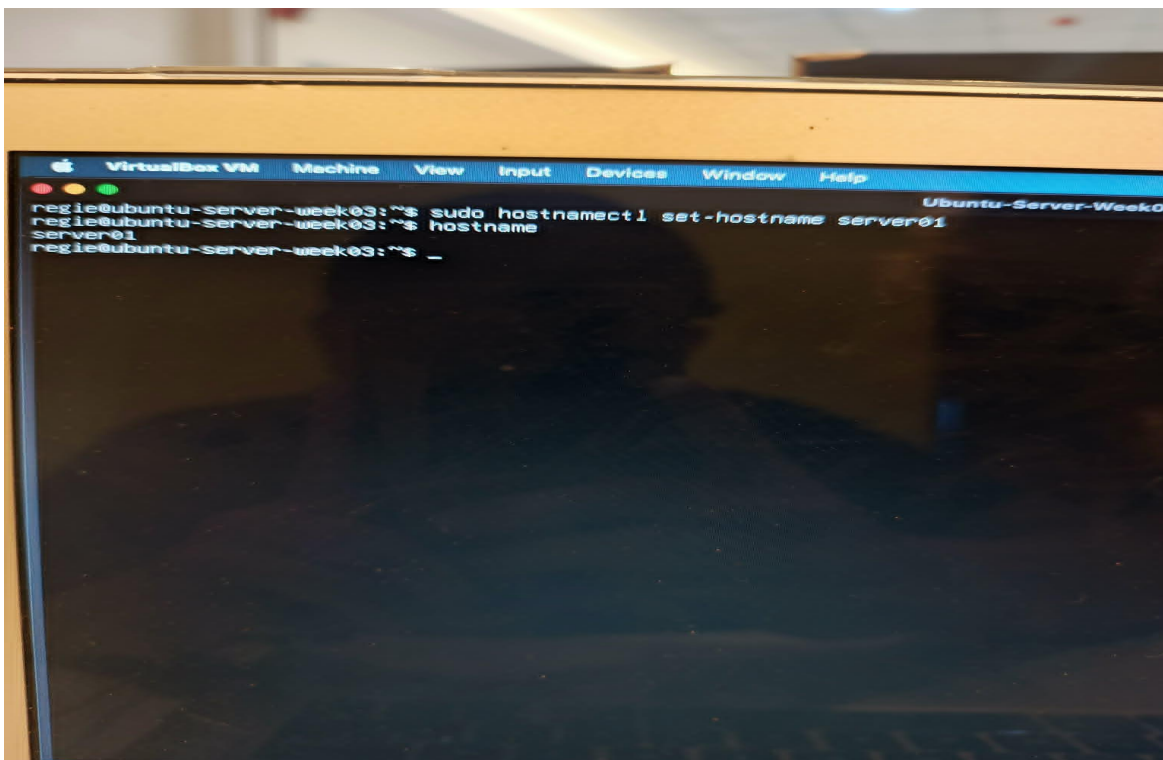
5. Server Verification

The following verification commands were completed after installation. The four screenshots available for the portfolio are documented below.

5.1 Hostname Verification

Command: hostname

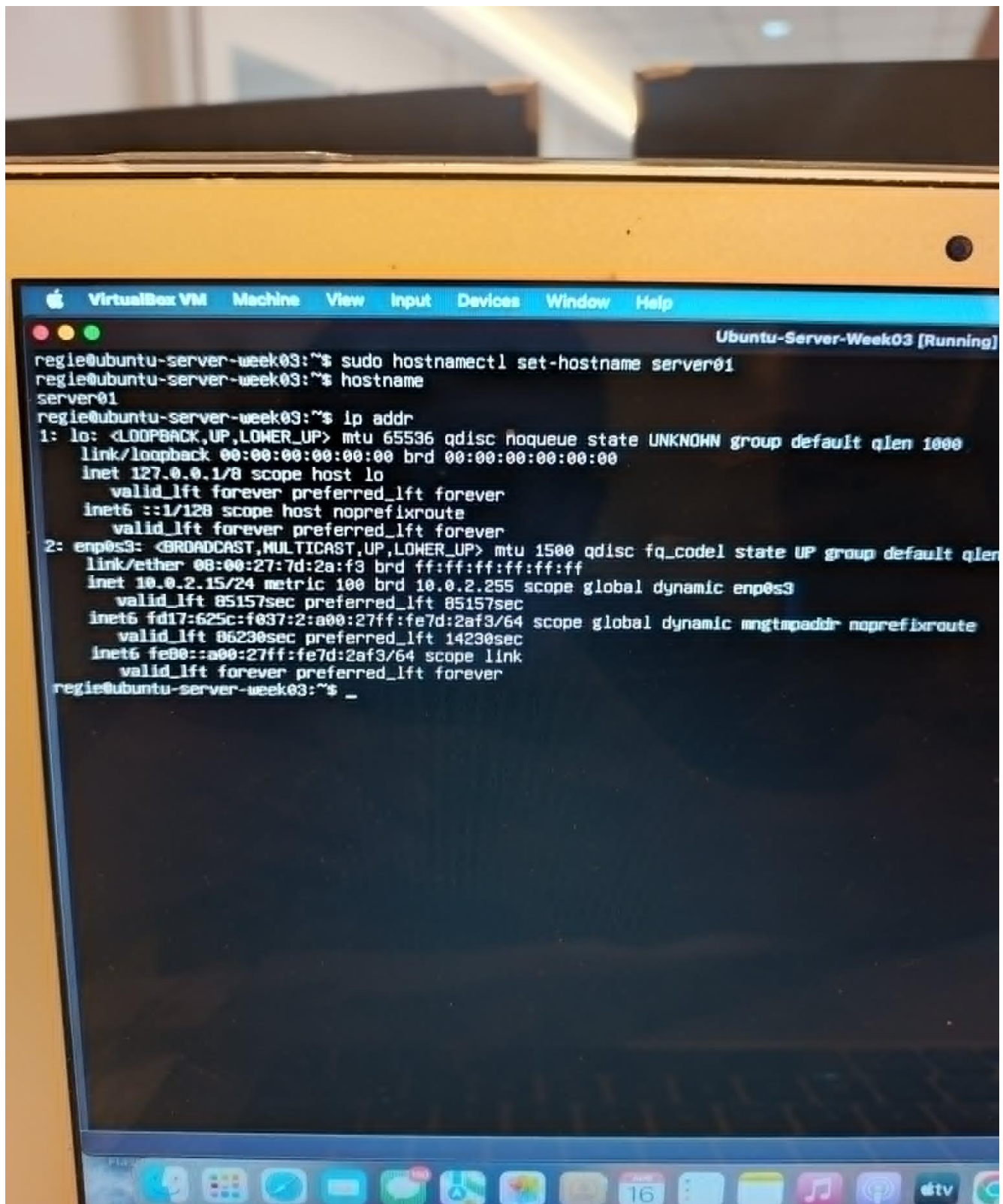
Expected/Observed Result: server01



5.2 IP Address Verification

Command: ip addr

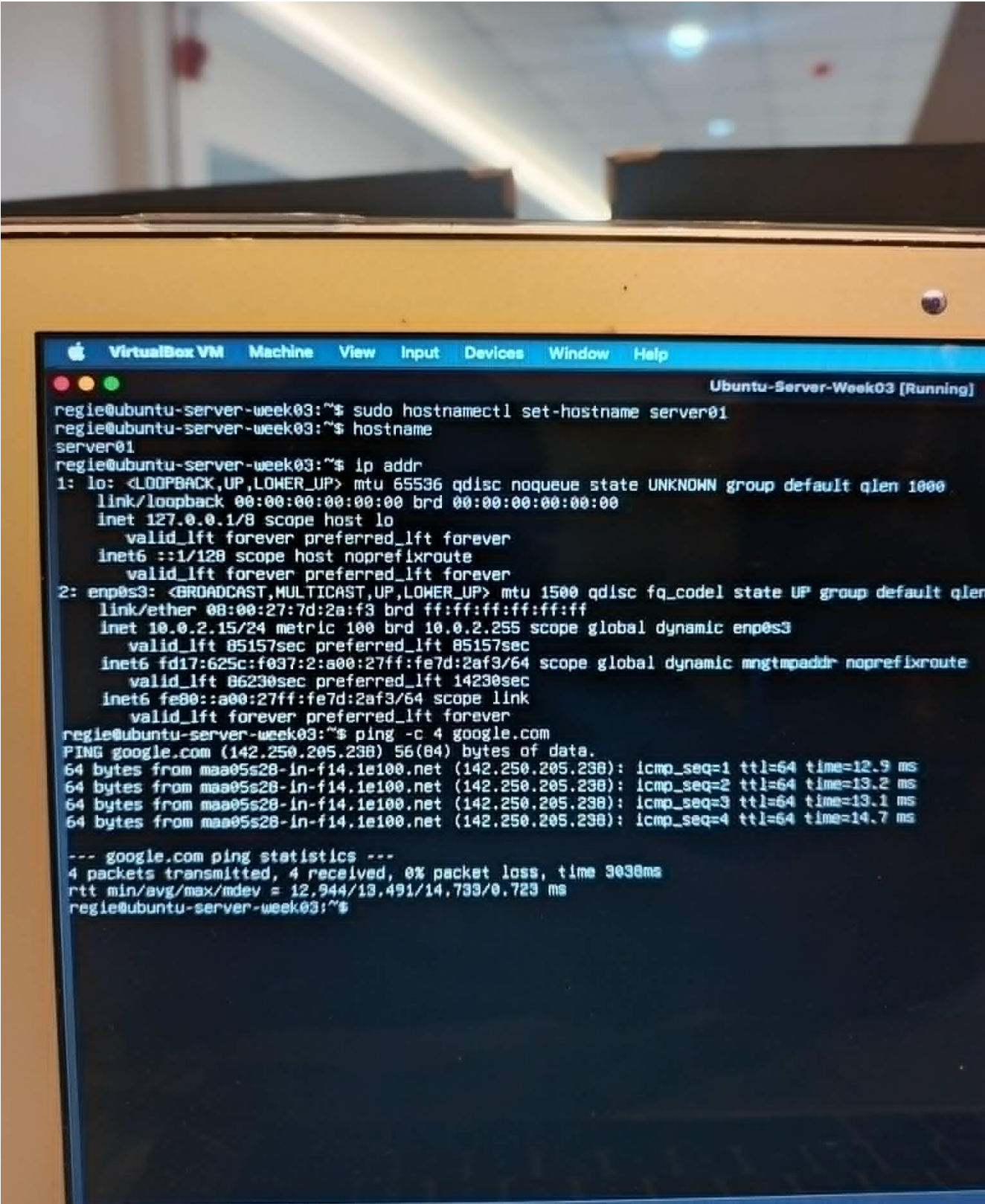
Expected/Observed Result: DHCP-assigned IPv4 address



5.3 Internet Connectivity

Command: `ping -c 4 google.com`

Expected/Observed Result: Successful replies / 0% packet loss



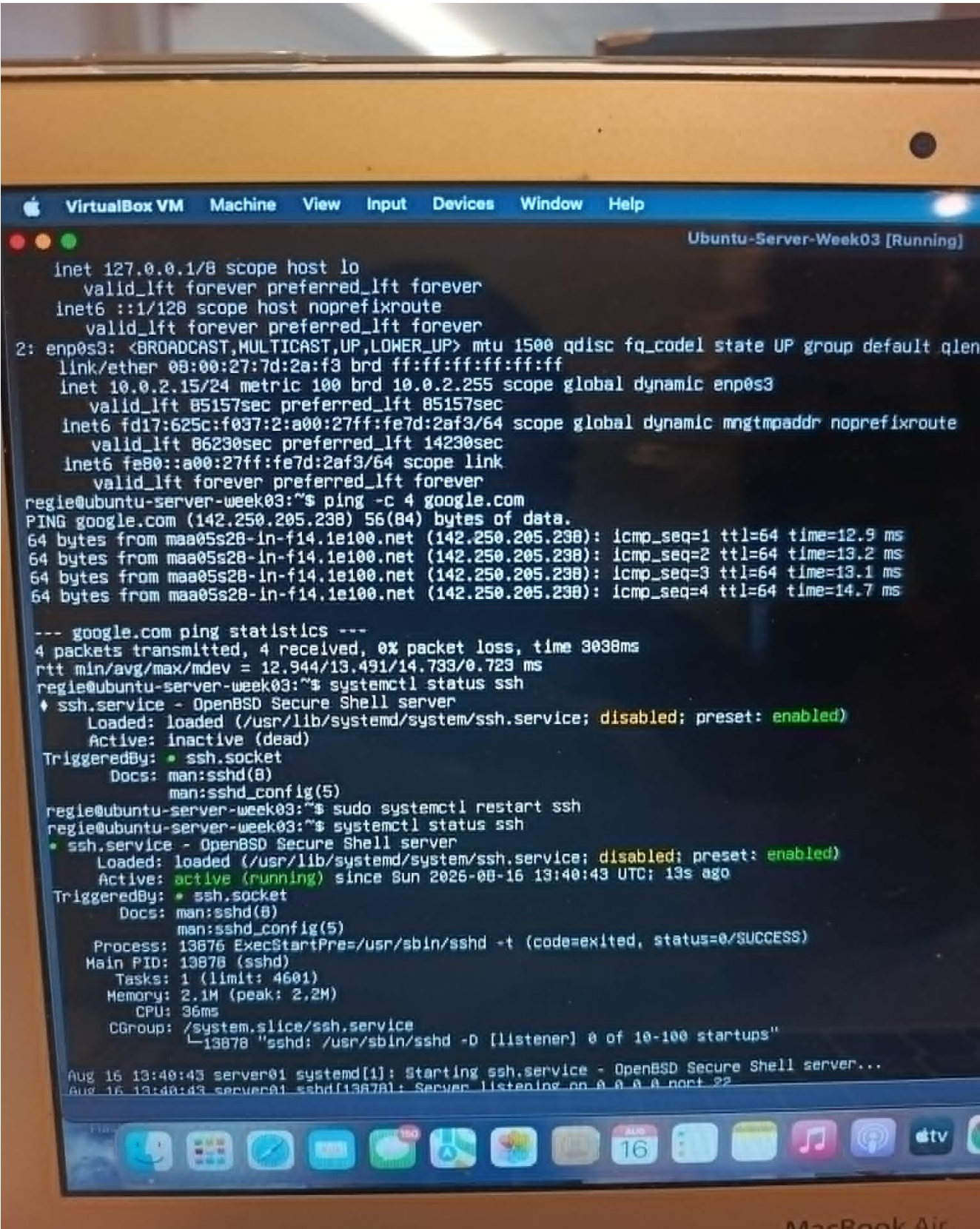
```
regie@ubuntu-server-week03:~$ sudo hostnamectl set-hostname server01
regie@ubuntu-server-week03:~$ hostname
server01
regie@ubuntu-server-week03:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
    link/ether 08:00:27:7d:2a:f3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85157sec preferred_lft 85157sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7d:2af3/64 scope global dynamic mngtppaddr noprefixroute
        valid_lft 86230sec preferred_lft 14230sec
    inet6 fe80::a00:27ff:fe7d:2af3/64 scope link
        valid_lft forever preferred_lft forever
regie@ubuntu-server-week03:~$ ping -c 4 google.com
PING google.com (142.250.205.238) 56(84) bytes of data:
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=1 ttl=64 time=12.9 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=2 ttl=64 time=13.2 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=3 ttl=64 time=13.1 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=4 ttl=64 time=14.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3030ms
rtt min/avg/max/mdev = 12.944/13.491/14.733/0.723 ms
regie@ubuntu-server-week03:~$
```

5.4 SSH Service Verification

Command: systemctl status ssh

Expected/Observed Result: Active: active (running)



```
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen
    link/ether 08:00:27:7d:2a:f3 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
        valid_lft 85157sec preferred_lft 85157sec
    inet6 fd17:625c:f037:2:a00:27ff:fe7d:2af3/64 scope global dynamic mngtmpaddr noprefixroute
        valid_lft 86230sec preferred_lft 14230sec
    inet6 fe80::a00:27ff:fe7d:2af3/64 scope link
        valid_lft forever preferred_lft forever
regie@ubuntu-server-week03:~$ ping -c 4 google.com
PING google.com (142.250.205.238) 56(84) bytes of data:
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=1 ttl=64 time=12.9 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=2 ttl=64 time=13.2 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=3 ttl=64 time=13.1 ms
64 bytes from maa05s28-in-f14.1e100.net (142.250.205.238): icmp_seq=4 ttl=64 time=14.7 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3038ms
rtt min/avg/max/mdev = 12.944/13.491/14.733/0.723 ms
regie@ubuntu-server-week03:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: inactive (dead)
TriggeredBy: ● ssh.socket
             Docs: man:sshd(8)
                  man:sshd_config(5)
regie@ubuntu-server-week03:~$ sudo systemctl restart ssh
regie@ubuntu-server-week03:~$ systemctl status ssh
* ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: enabled)
   Active: active (running) since Sun 2025-08-16 13:40:43 UTC; 13s ago
TriggeredBy: ● ssh.socket
             Docs: man:sshd(8)
                  man:sshd_config(5)
   Process: 13676 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 13878 (sshd)
     Tasks: 1 (limit: 4601)
    Memory: 2.1M (peak: 2.2M)
       CPU: 36ms
    CGroup: /system.slice/ssh.service
            └─13878 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Aug 16 13:40:43 server01 systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Aug 16 13:40:43 server01 sshd[13878]: Server listening on 0.0.0.0 port 22
```

6. System Update

The server was updated using the required commands:

sudo apt update

sudo apt upgrade -y

The update and upgrade completed successfully. The terminal reported that packages were available for upgrade and then returned to the normal shell prompt.

7. BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	Legacy firmware used to initialize hardware and start the OS.	Modern standardized firmware interface for initialization and boot applications.
Boot Process	POST → boot device → MBR boot code → boot loader.	Firmware initialization → EFI boot application/boot loader → OS.
Maximum Disk Support	Traditionally limited by MBR to about 2 TB.	Works with GPT and supports much larger disks.
Partition Style	MBR	GPT
Security	Limited compared with modern UEFI.	Supports Secure Boot and other modern security capabilities.
Speed	Less flexible on modern hardware.	Designed for efficient, flexible modern booting.
Advantages	Simple and compatible with legacy systems.	Large-disk support, GPT, Secure Boot, more partitions, extensibility.
Disadvantages	Legacy storage and security limitations.	More complex; older systems may not support it.
Modern Usage	Primarily legacy systems.	Standard for most modern computers.

Why UEFI Has Largely Replaced BIOS

BIOS is the traditional firmware technology used to initialize hardware and start an operating system. It commonly relies on the Master Boot Record (MBR), which has limitations such as the traditional approximately 2 TB disk-size limit and fewer partitioning capabilities.

UEFI provides a more modern and extensible pre-boot environment. It works well with the GUID Partition Table (GPT), which supports much larger storage devices and many more partitions. UEFI also supports technologies such as Secure Boot, helping protect the boot process from unauthorized boot software.

Modern computers require larger storage, stronger security, and more flexible boot management. For these reasons, UEFI has largely replaced legacy BIOS. BIOS knowledge remains useful when administering older computers and legacy systems.

8. Ubuntu Boot Process Flowchart

Create the flowchart with the following sequence:

Power On → BIOS/UEFI Initialization → Boot Device Detection → Boot Loader (GRUB) → Linux Kernel → init/systemd → Services Start → Login Prompt

Export the completed flowchart as PNG and PDF for the portfolio.

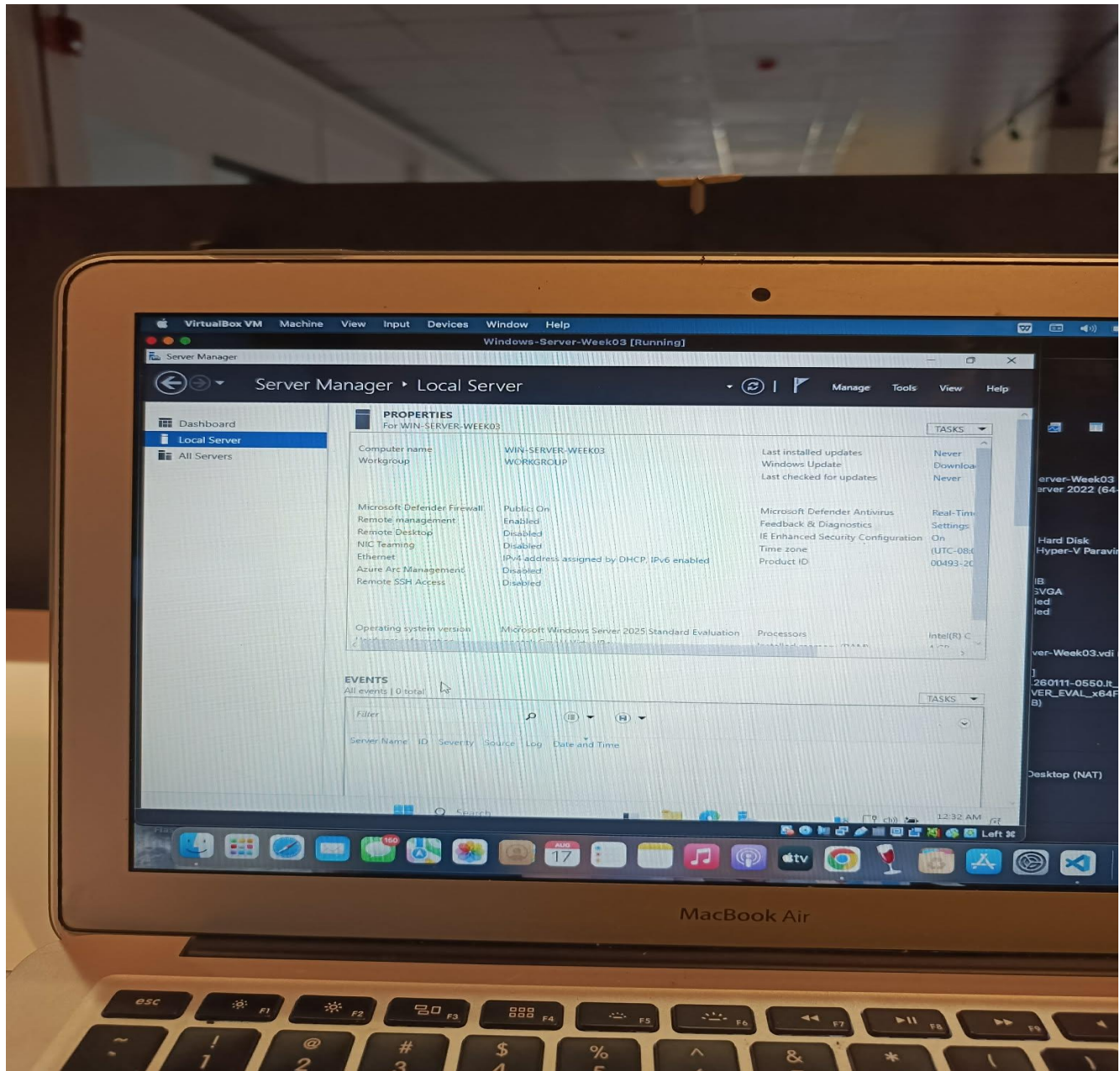
Ubuntu Boot Process Flowchart

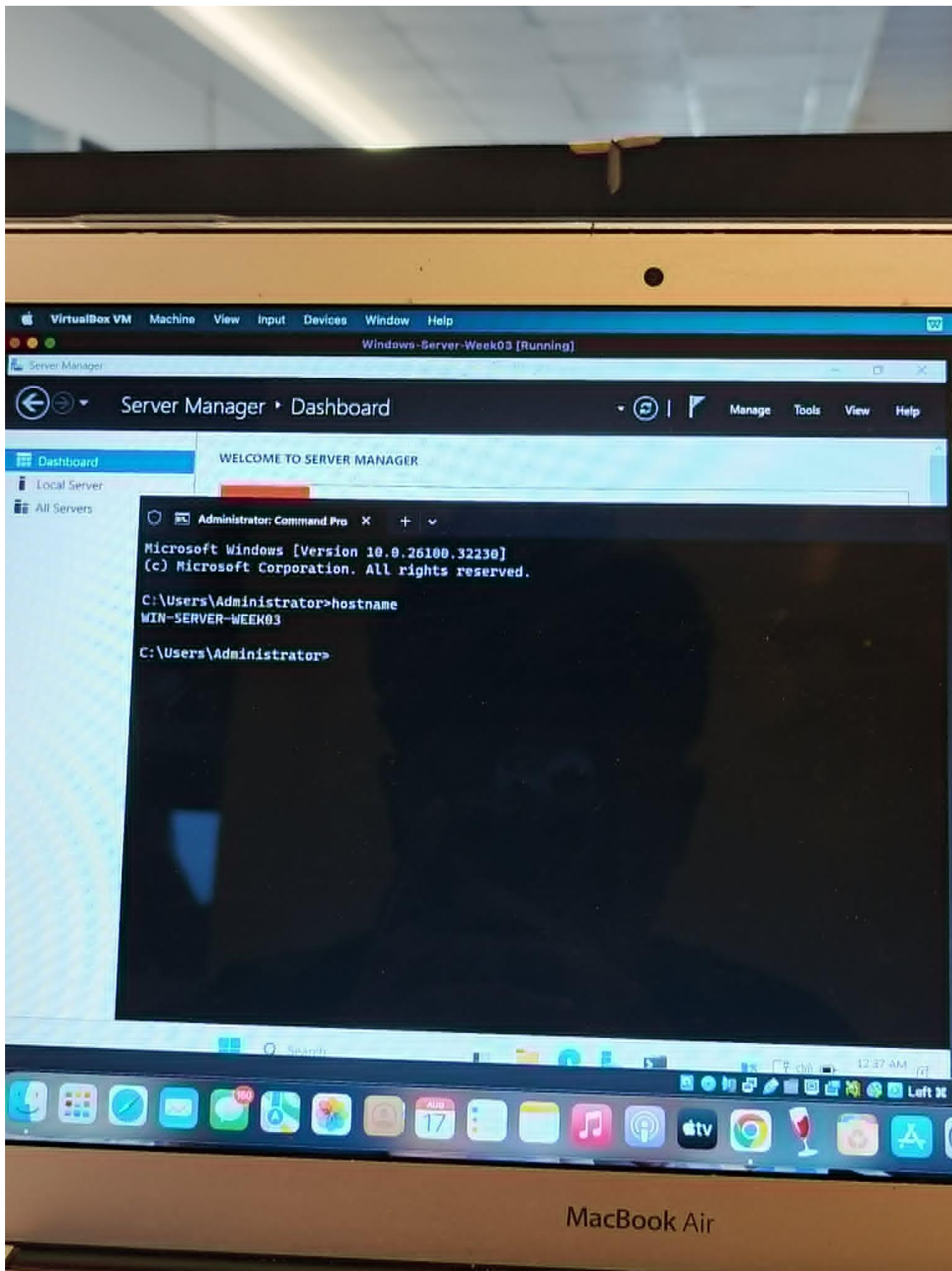
Regie M. Estravela



9. Windows Server Evaluation – Bring-Home Activity

Install Windows Server Evaluation in a separate virtual machine. Assign a computer name, create an administrator password, log in successfully, and capture installation screenshots.





10. Operating System Comparison

Category	Windows Server	Ubuntu Server	Rocky Linux
Licensing	Proprietary/commercial licensing; evaluation editions available.	Open-source with optional commercial support.	Open-source enterprise Linux distribution compatible with the RHEL ecosystem.
User Interface	GUI and command-line administration.	Primarily command-line for server installations.	Primarily command-line for server installations.
Package Management	Windows Update and Windows package-management tools.	APT / dpkg.	DNF / RPM.
Security	Active Directory, Group Policy, Defender, security baselines.	Linux permissions, sudo, SSH, firewall tools, AppArmor.	SELinux, Linux permissions, firewalld, SSH.
Performance	Strong for Microsoft workloads and enterprise services.	Efficient and flexible for web, cloud, databases, and infrastructure.	Stable platform for enterprise Linux workloads.
Typical Enterprise Use	Active Directory, Microsoft services, Windows applications, file services.	Web servers, cloud, databases, containers, internal services.	Enterprise Linux servers and RHEL-compatible workloads.
Advantages	Excellent Microsoft ecosystem integration and centralized management.	Free, flexible, broad community and software ecosystem.	Stable, enterprise-oriented, RHEL-compatible.
Disadvantages	Licensing costs and Microsoft ecosystem dependence.	Some software is more Windows-oriented.	Requires Linux administration knowledge and has a smaller general-purpose community.

11. Problems Encountered and Solutions

Ubuntu installer remained at Reading package lists

Solution: The archive mirror was adjusted and the installation was allowed to continue.

VM returned to the installer after completion

Solution: The Ubuntu ISO was removed from the virtual optical drive before rebooting.

Hostname initially did not match the module

Solution: The hostname was changed to server01 using hostnamectl.

SSH status needed verification

Solution: The SSH service was restarted and verified until it displayed active (running).

12. Lessons Learned

- Correct VM specifications and installation settings are important for reliable server deployment.
- Hostname, networking, storage, user accounts, and SSH are essential initial server configuration tasks.
- Installation media should be detached after installation so the VM boots from the installed disk.
- Verification commands provide evidence that the server is ready for use.

- Professional screenshots and documentation make server deployments reproducible.

13. Professional Documentation Checklist

- ☐ Four server verification screenshots inserted
- ☐ BIOS vs UEFI comparison completed
- ☐ Boot Process Flowchart exported as PNG and PDF
- ☐ Windows Server Evaluation completed
- ☐ Windows/Ubuntu/Rocky Linux comparison completed
- ☐ Problems and solutions documented
- ☐ Lessons learned documented
- ☐ Installation Guide exported as PDF
- ☐ Professional Installation Manual created
- ☐ README.md completed
- ☐ GitHub repository updated
- ☐ LinkedIn reflection posted

14. References

- Ubuntu Server Documentation — <https://documentation.ubuntu.com/server/>
- UEFI Forum — <https://uefi.org/>
- Microsoft Learn — <https://learn.microsoft.com/>
- Rocky Linux Documentation — <https://docs.rockylinux.org/>